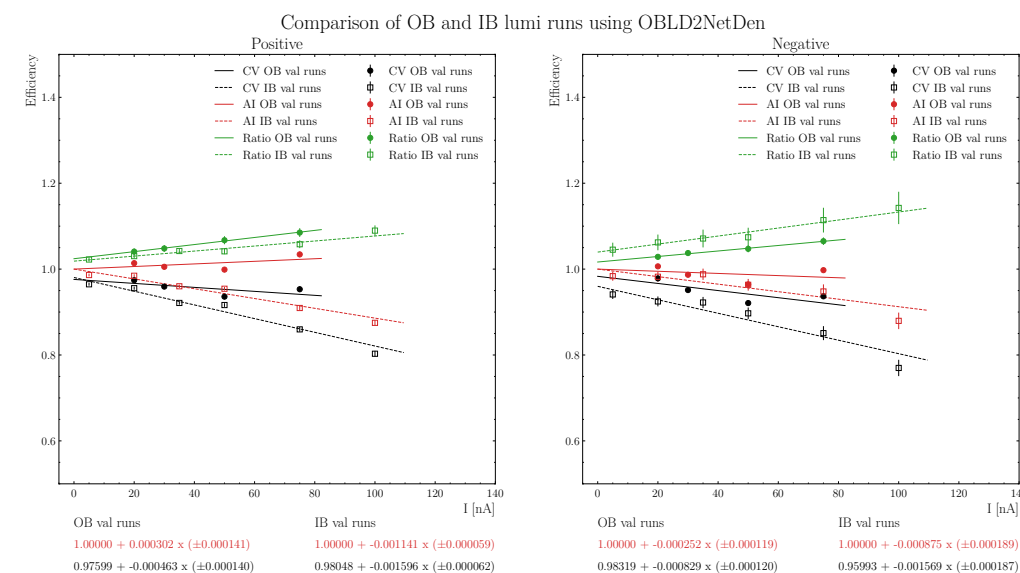


Analysis group meeting

Mathieu Ouillon

6/2/25

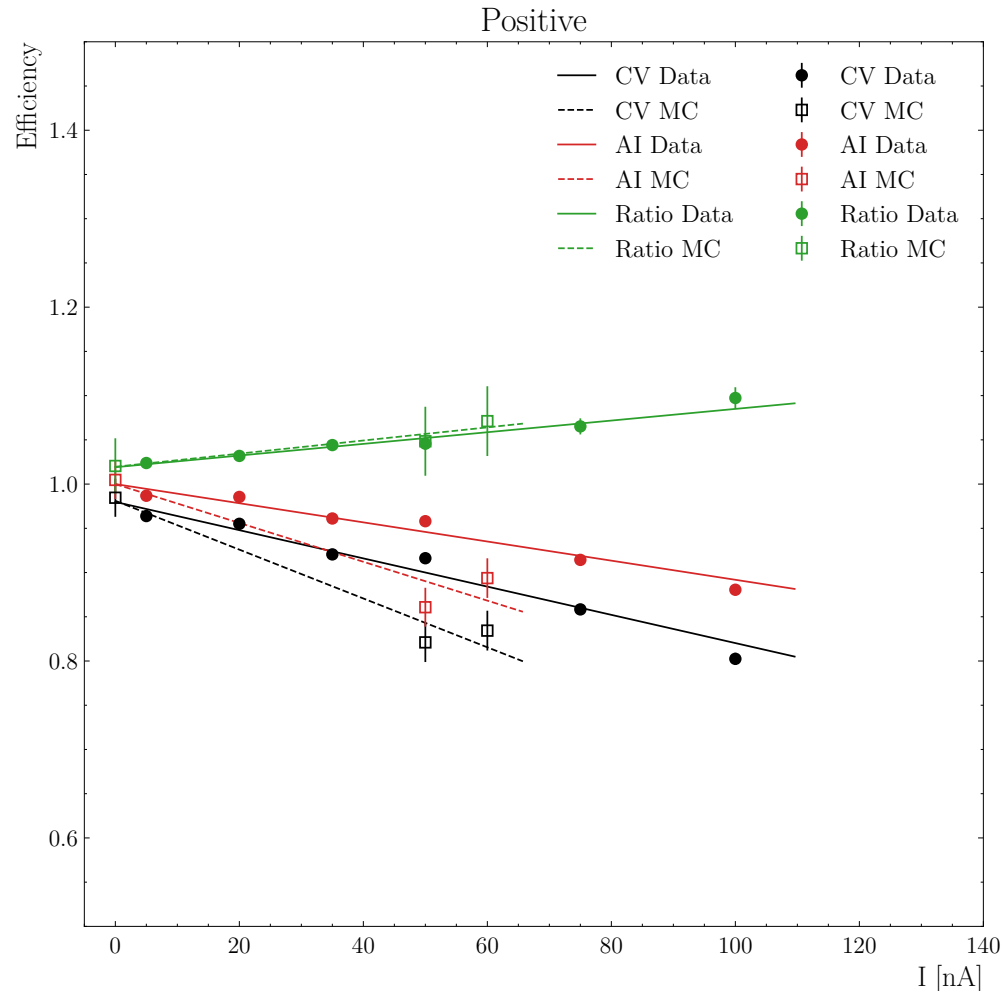
- Cooking for AI network:
 - All training cooking are done (OBLD2NoDen, IBLD2NoDen, IBLD2Den , IBLD2NoDen)
 - All validation cooking are done ()
- For the MC vs data study:
 - All the data cooking is done for LD2 (OBLD2NetDen, OBLD2NetNoDen)
 - MC cooking only with denoising is done; no denoising is in progress
- I have checked the OB LD2 Net on OB and IB LD2. I still have a positive slope for OB for AI positive particle.



MC vs data study

- SIDIS events with LD2 target, GEMC 5.11, COATJAVA 12.0.4t, timestamp 03/28/25-12:00:00 (same as pass0v10), network: OBLD2Net+Den. Simulation 0nA, 50nA and 60nA

Comparison of MC vs. Data with denoising using OBLD2NetDen



Data

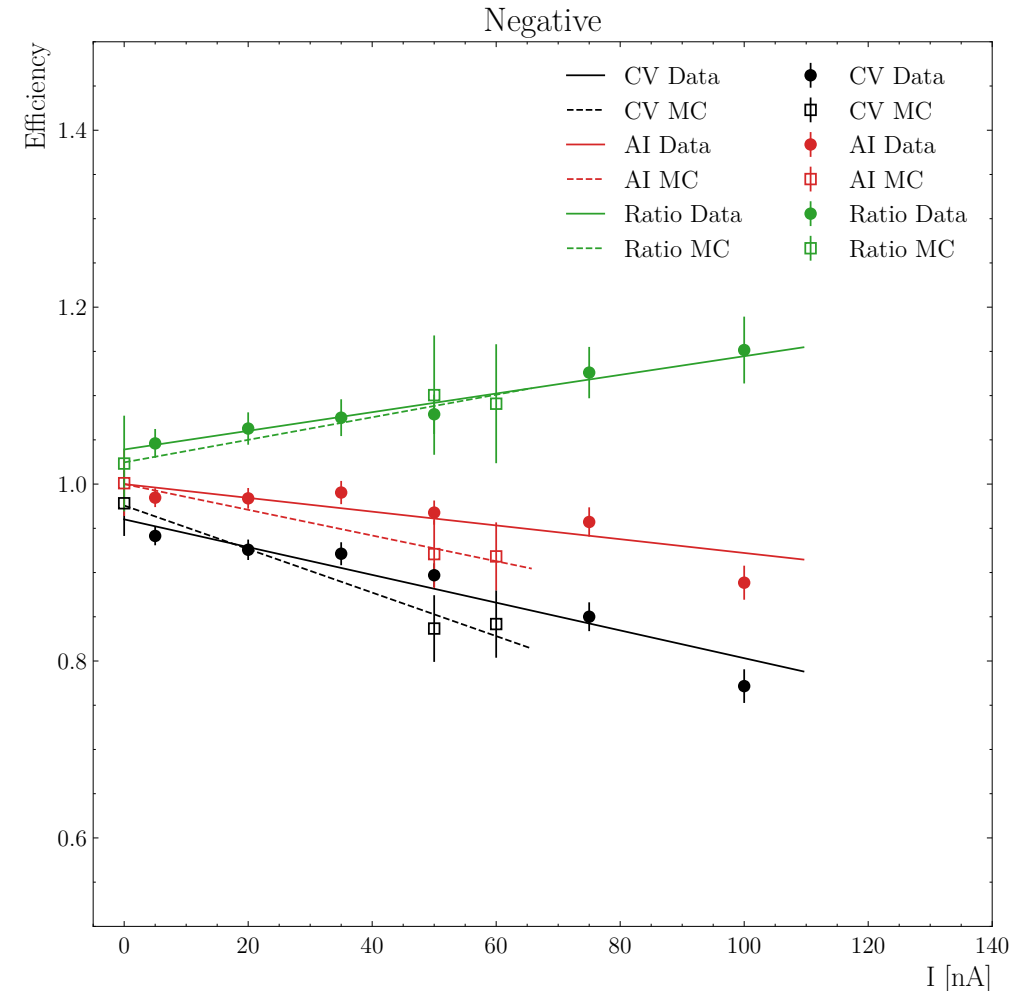
$$1.00000 + -0.001082 \times (\pm 0.000059)$$

$$0.97983 + -0.001596 \times (\pm 0.000062)$$

MC

$$1.00000 + -0.002196 \times (\pm 0.000480)$$

$$0.98111 + -0.002760 \times (\pm 0.000481)$$



Data

$$1.00000 + -0.000779 \times (\pm 0.000189)$$

$$0.96013 + -0.001569 \times (\pm 0.000187)$$

MC

$$1.00000 + -0.001453 \times (\pm 0.000826)$$

$$0.97574 + -0.002460 \times (\pm 0.000821)$$

- Important difference but
- Use the wrong variation in the yaml file: `rgd_fall2023` instead of `rgd_fall2023_mc`
- Next step:
 - More statistics
 - network: OBLD2Net+NoDen
 - On CuSn target: Should I generate 50% of Cu events and 50% Sn?

- Run 19038 Information:
 - from logbook: HWP IN
 - from [Maya database](#): HWP IN
 - from CCDB: HWP OUT

11:17 Stop run 19037 with 100M events, start **19038**
 11:30 BMT HV SEC2 L4 OFF/ON.
 11:55 Stop run **19038** with 33M, changing the HWP to OUT (Hall-C request).

Run range 19038 - 19199
 Author baltzell
 Exact id /runcontrol/hwp:19038:default:2024-12-06_11-22-10
[\[Download text data\]](#)

Comment:

Data:

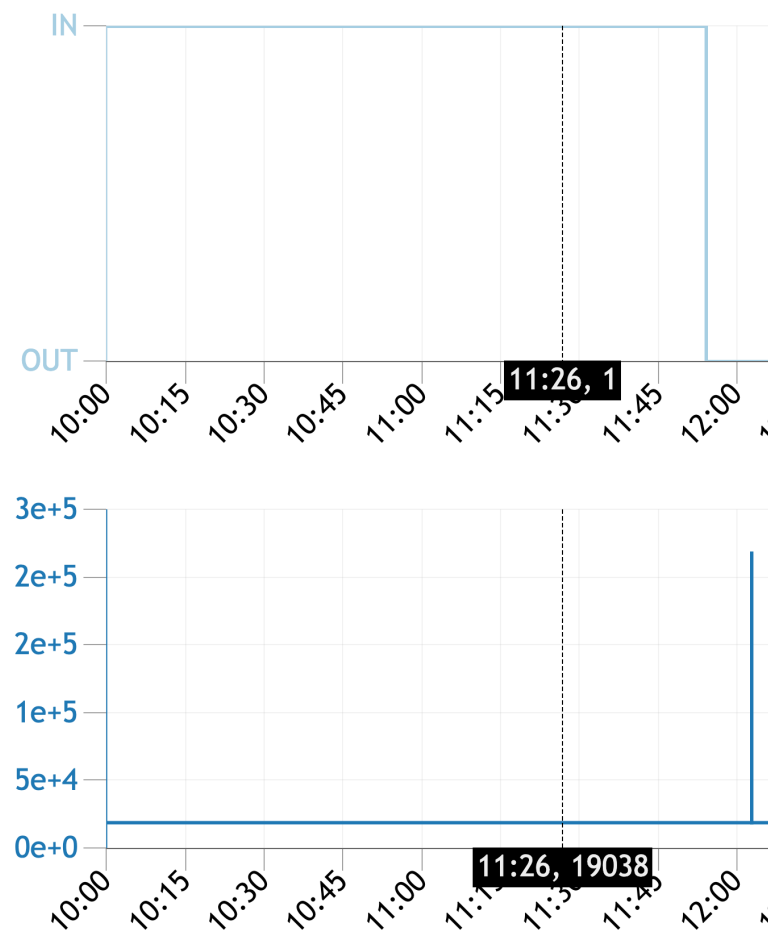
Show entries

sector	layer	component	hwp
0	0	0	1

Showing 1 to 1 of 1 entries

Search:

First Previous **1** Next Last

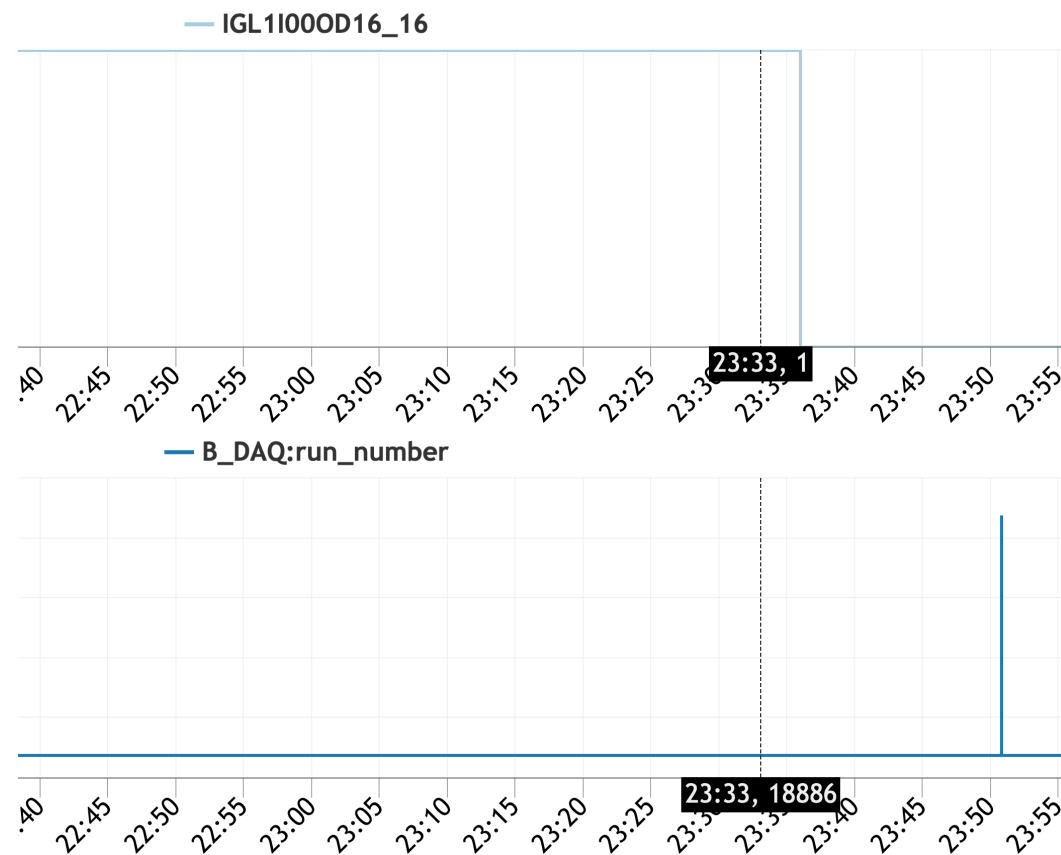


- Run 18886 Information:
 - from logbook: HWP IN
 - from [Maya database](#): HWP IN
 - from CCDB: HWP OUT

22:57- Started Run **18886**

23:33- Ended run **18886** for half wave plate change

November 21 - 22, 2023 (21:59:55 - 00:04:55)



Run range 18886 - 18956
 Author baltzell
 Exact id /runcontrol/hwp:18886:default:2024-12-06_11-22-09
[\[Download text data\]](#)

Comment:

Data:

Show entries

sector	layer	component	hwp
0	0	0	1

Showing 1 to 1 of 1 entries

Search:

First Previous **1** Next Last

• Run 18831 Information:

- from logbook: HWP OUT
- from [Maya database](#): HWP OUT
- from CCDB: HWP IN

13:06 roc_reboot successful; restarted DAQ from configure and taking run **18831** now
 13:29 Hall C is putting the HWP in, end run **18831** with 15m events. Hall C ran flipper script: <https://logbooks.jlab.org/entry/4222333>

Run range 18829 - 18885
 Author baltzell
 Exact id /runcontrol/hwp:18829:default:2024-12-06_11-22-09
[\[Download text data\]](#)

Comment:

Data:

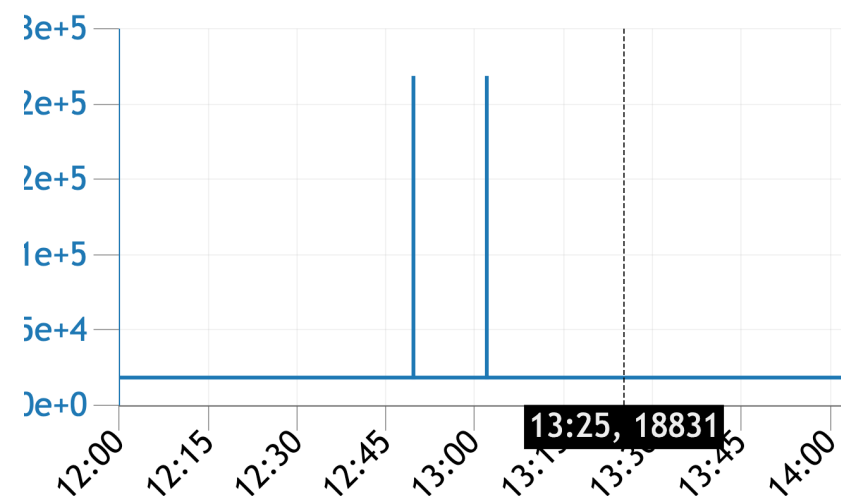
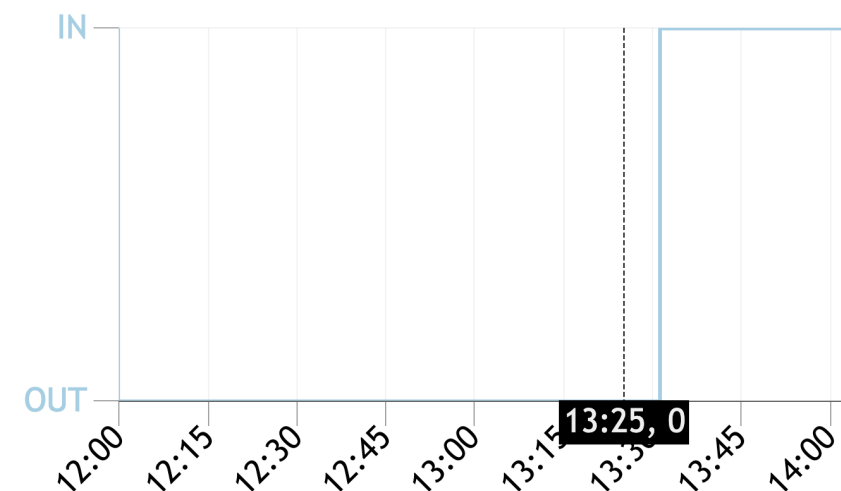
Show entries

sector	layer	component	hwp
0	0	0	-1

Showing 1 to 1 of 1 entries

Search:

First Previous **1** Next Last

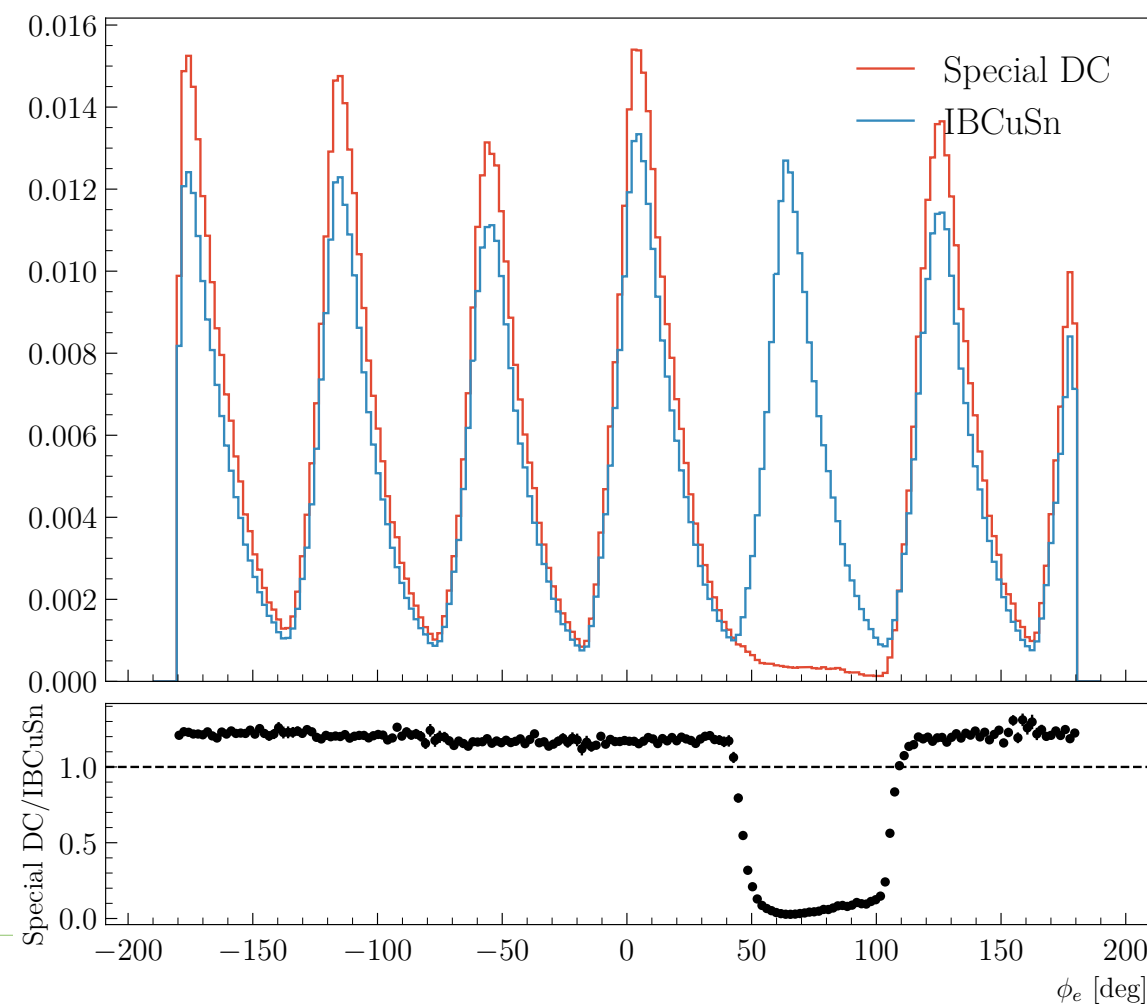
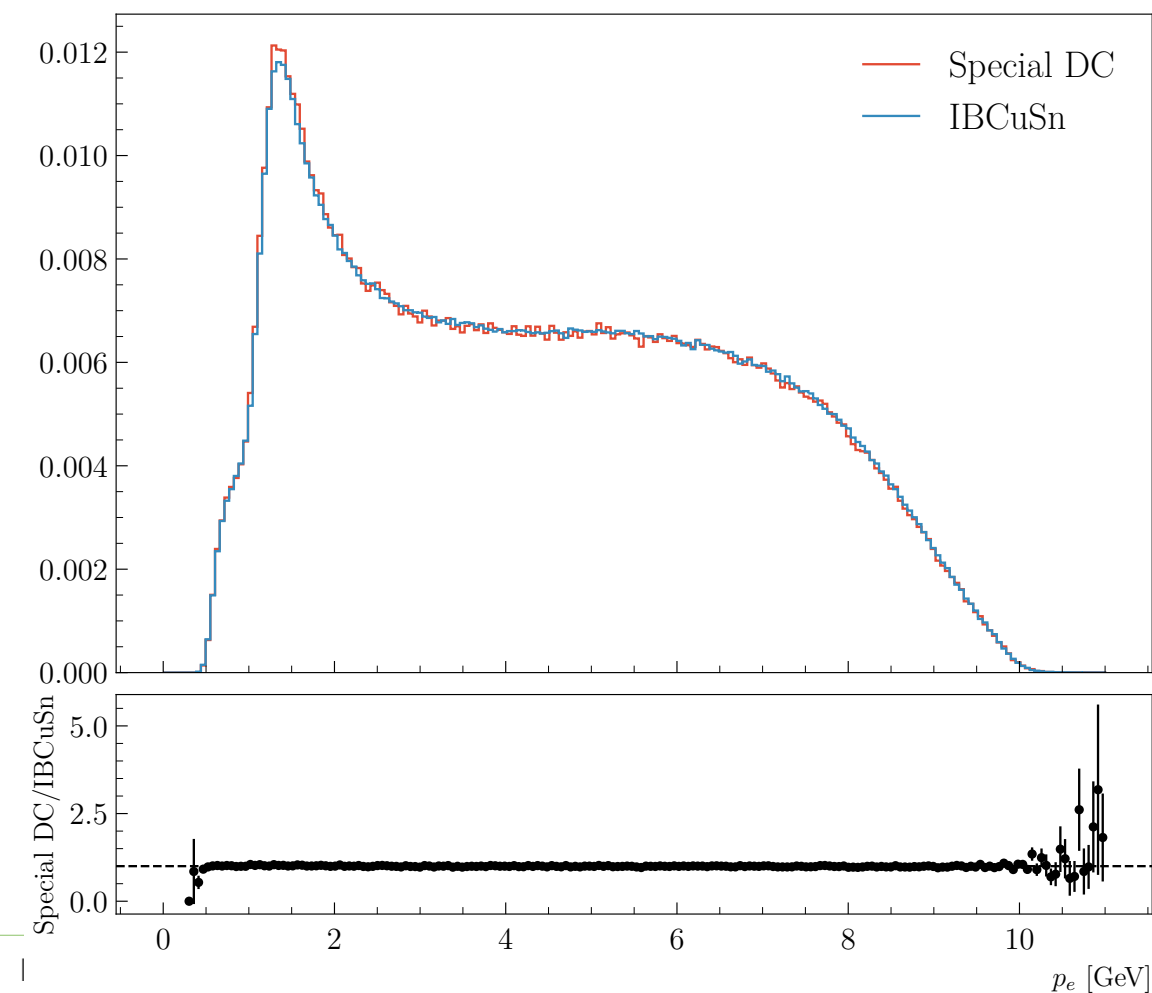


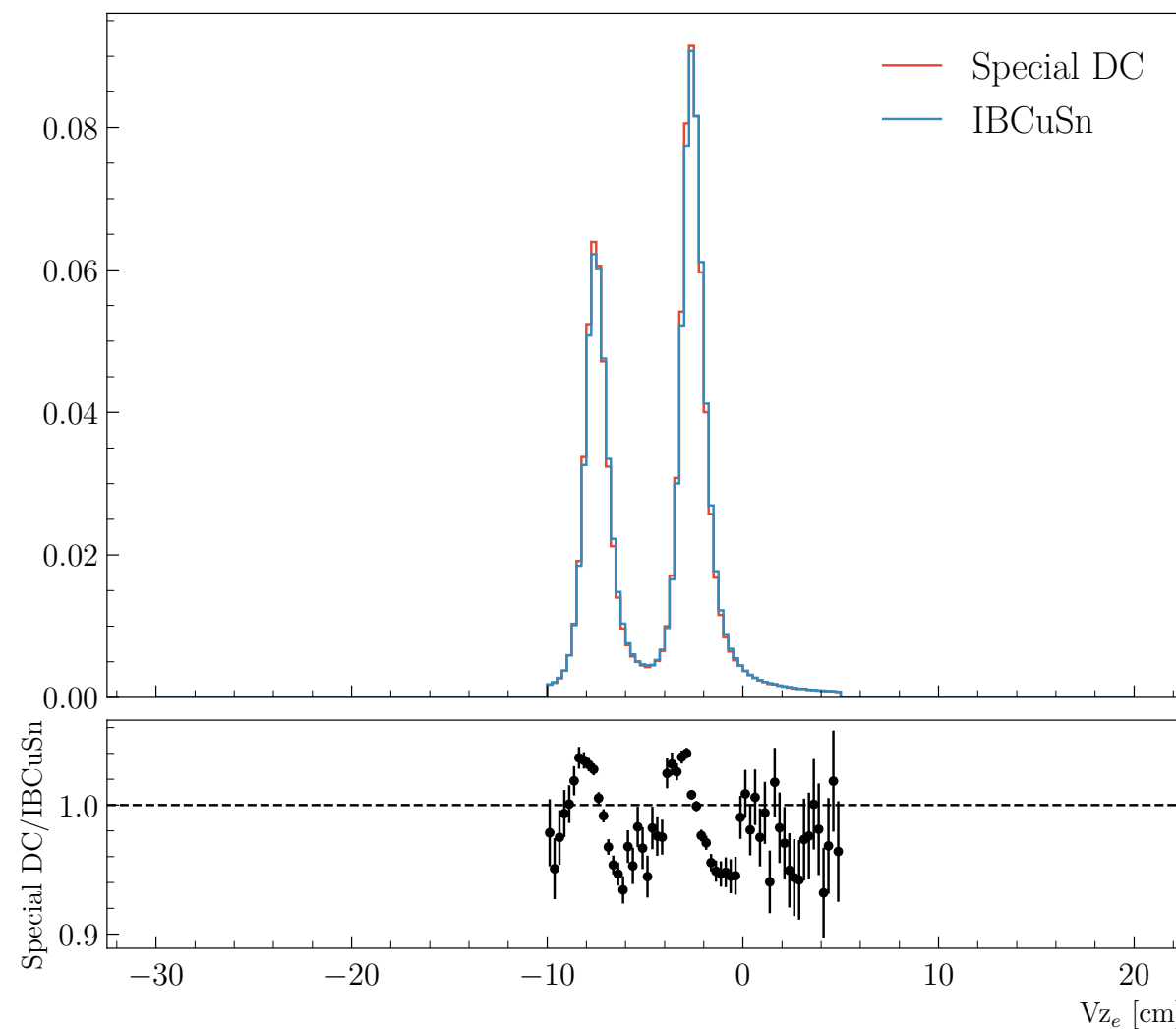
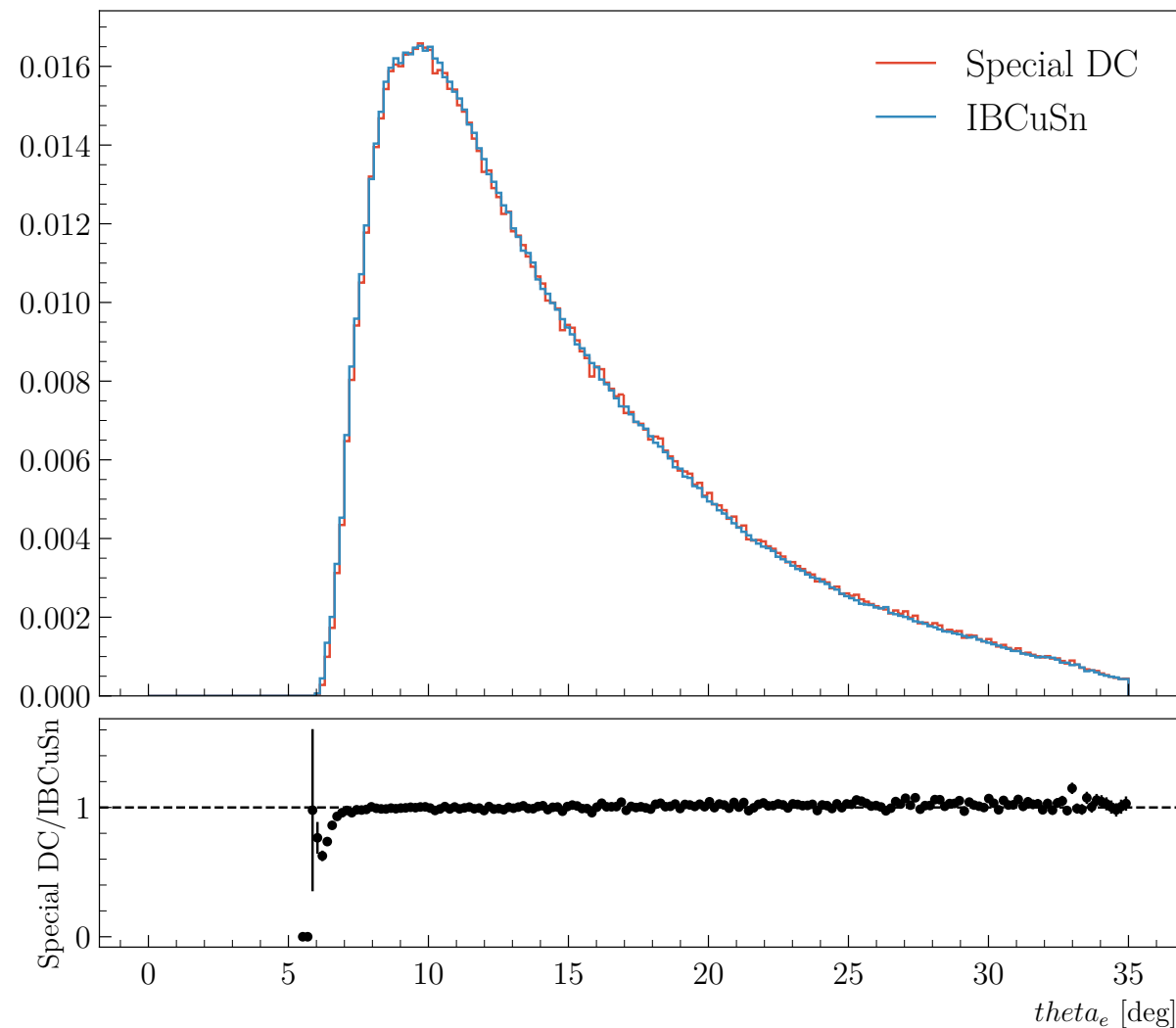
Physics quantities with special DC runs

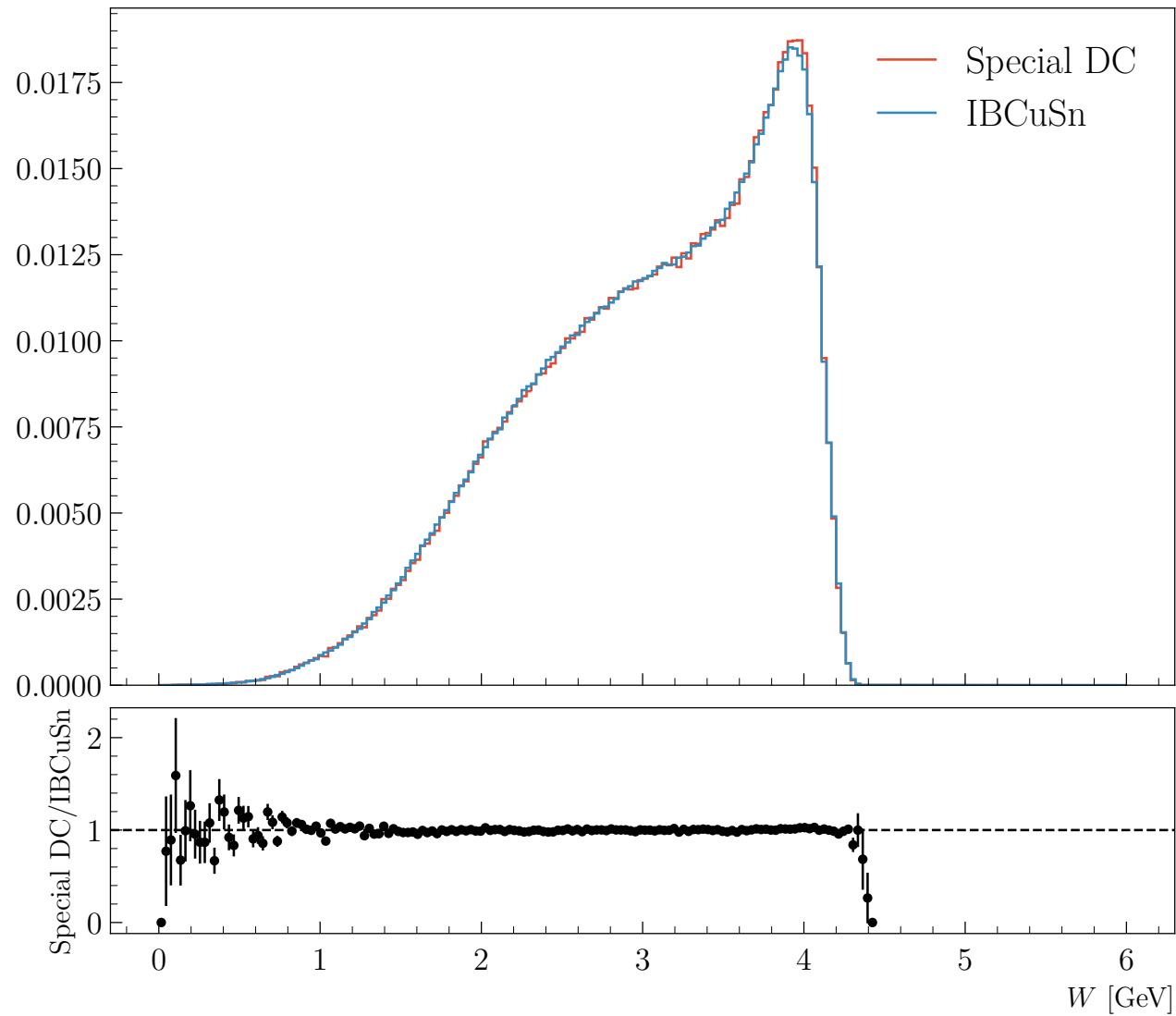
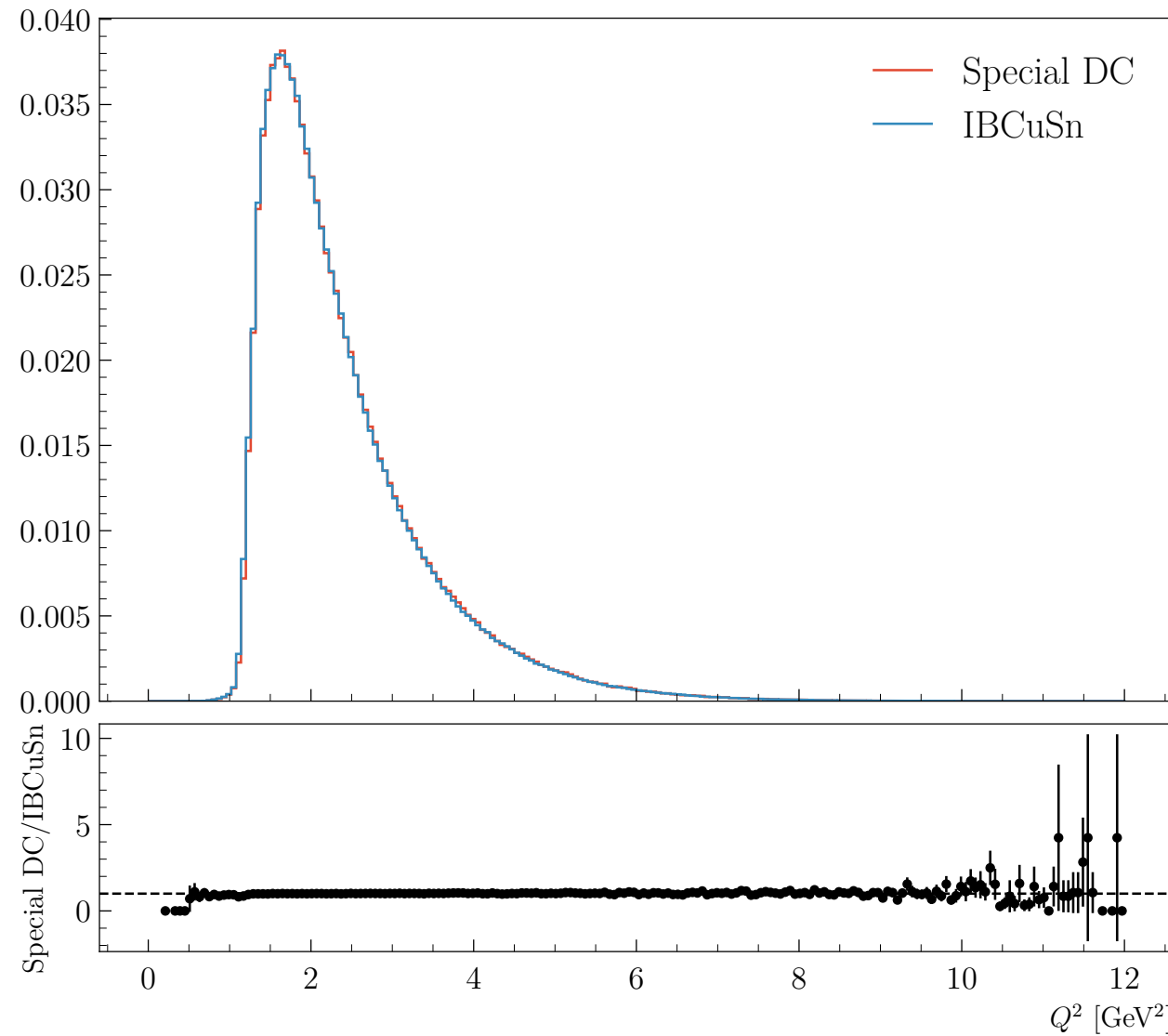
8

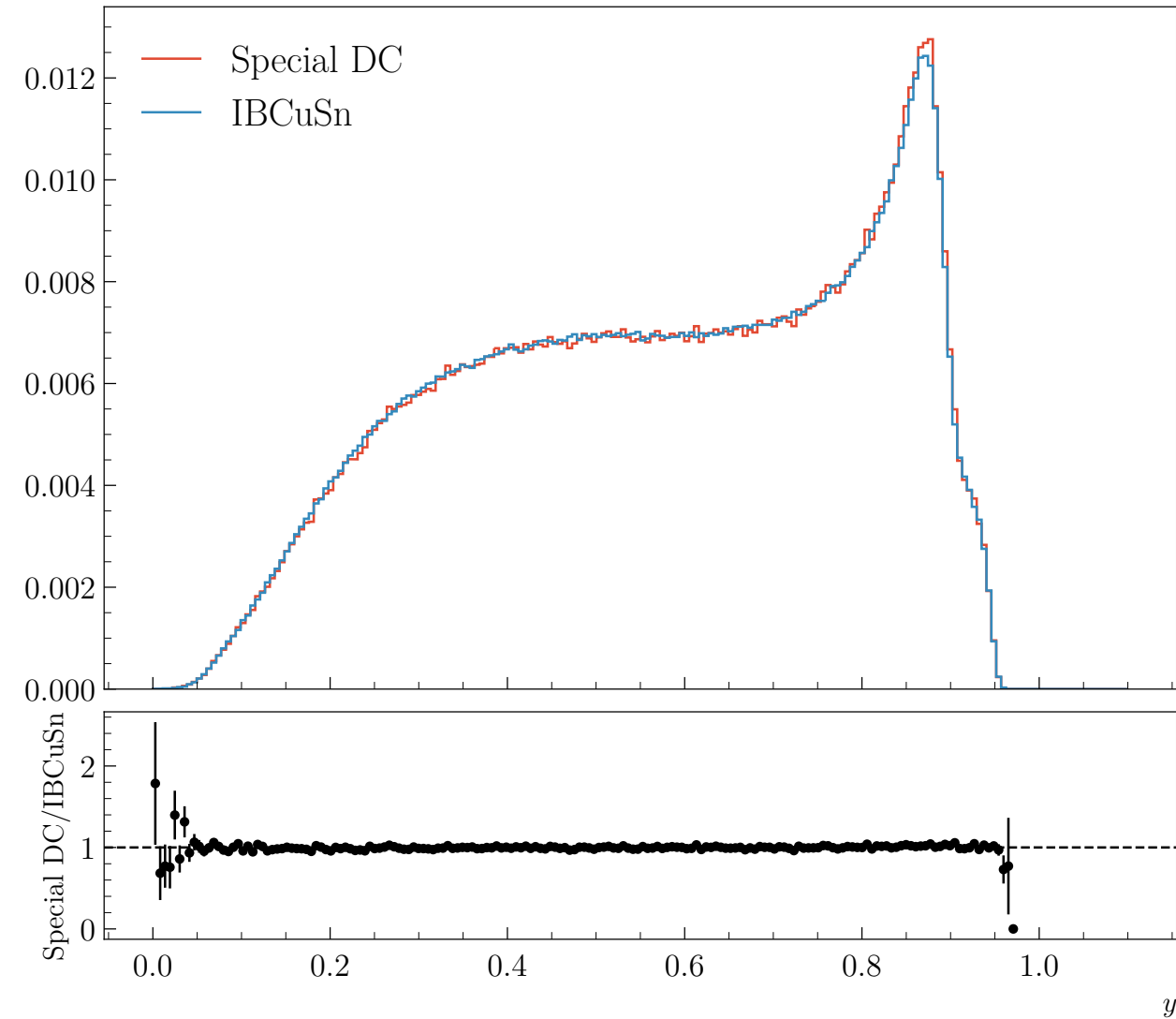
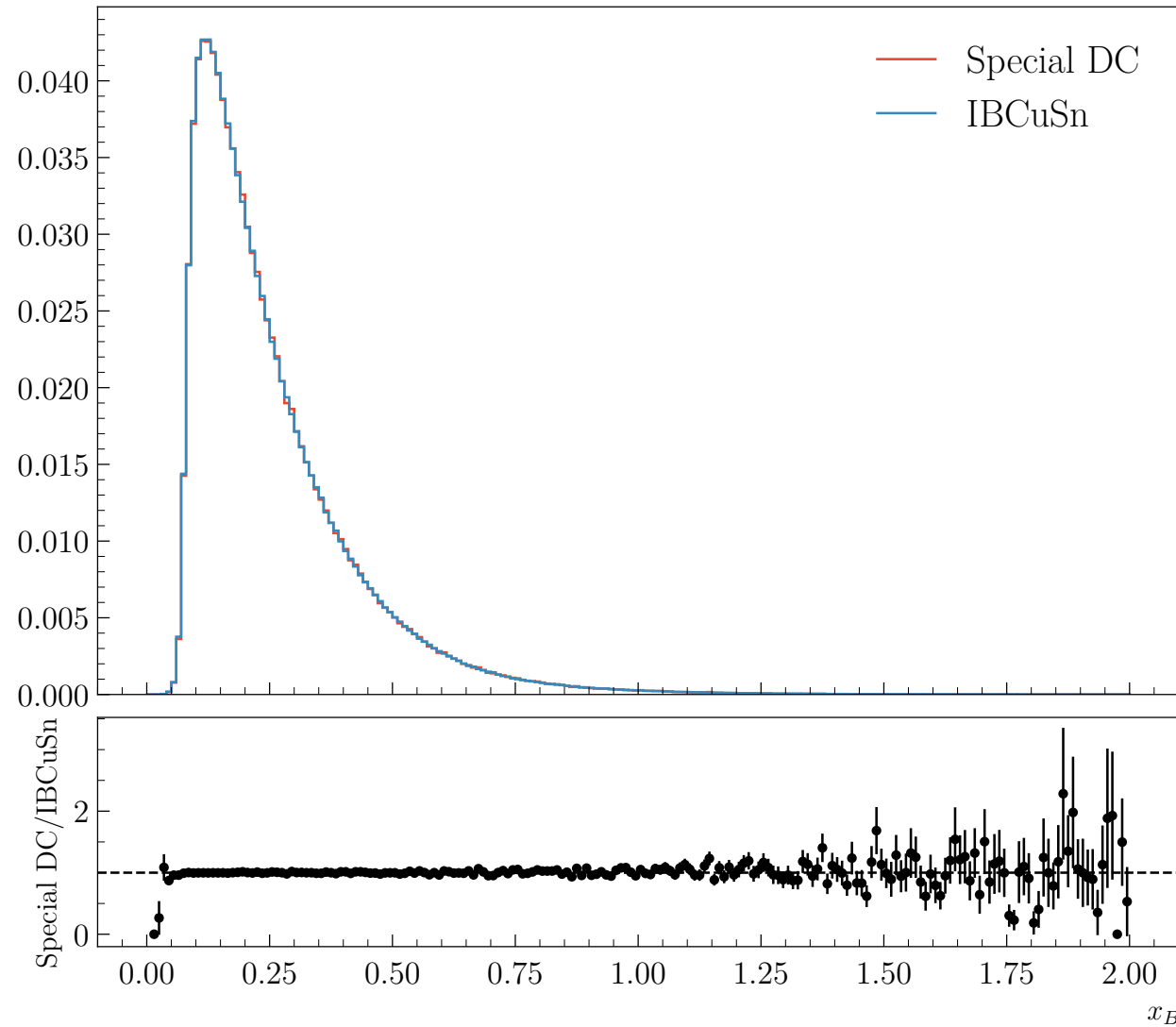
Look at the CuSn runs without DC Sector 2: 18360, 18361, 18362, 18363, 18364, 18365, 18366, 18368

Compared to inbending CuSn Runs, electron variables









Fiducial cuts on DC for electron

- Following the work done in the [RG-M common analysis note](#).
- Try to implement DC fiducial cut for the electron using the edge variable

RG-M Analysis Note: 6 GeV electron proton selection and Particle ID

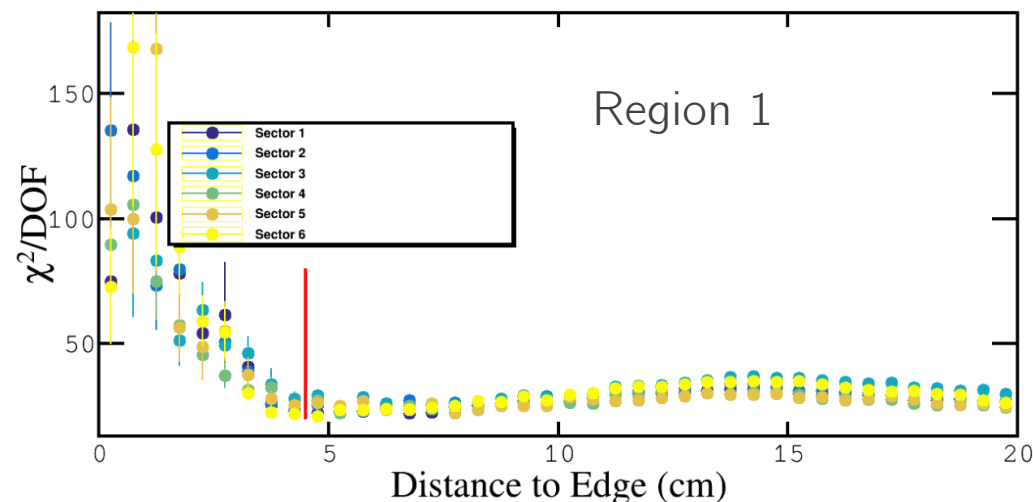
```
{"name":"edge", "type":"F", "info":"signed distance between track position and closest detector edge (cm)"}
```

From the REC::Traj bank and chi2/ndf

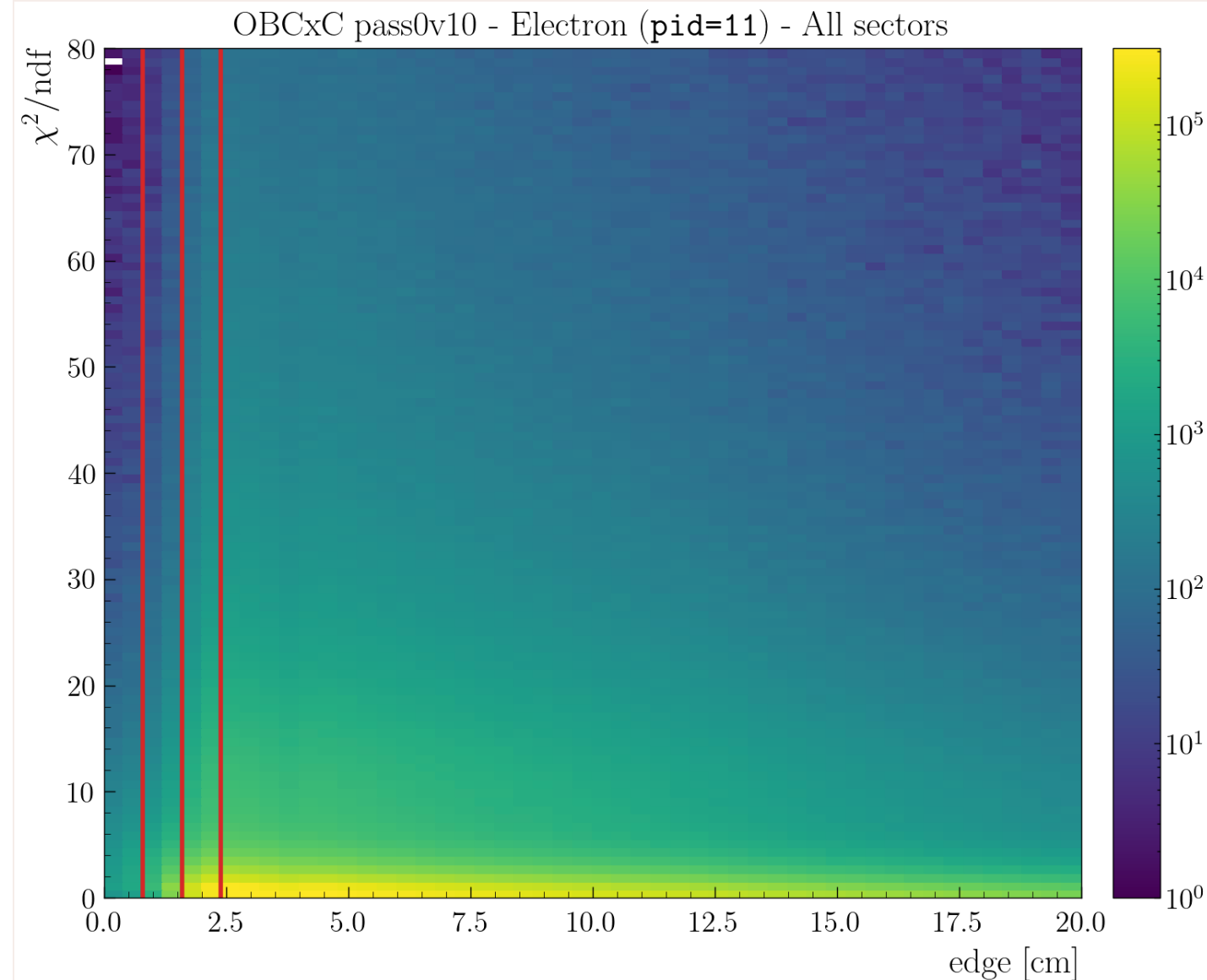
```
{"name":"chi2", "type":"F", "info":"Chi2 (or quality) track fitting"},  
{"name":"NDF", "type":"S", "info":"number of degrees of freedom in track fitting"},
```

From the REC::Track bank

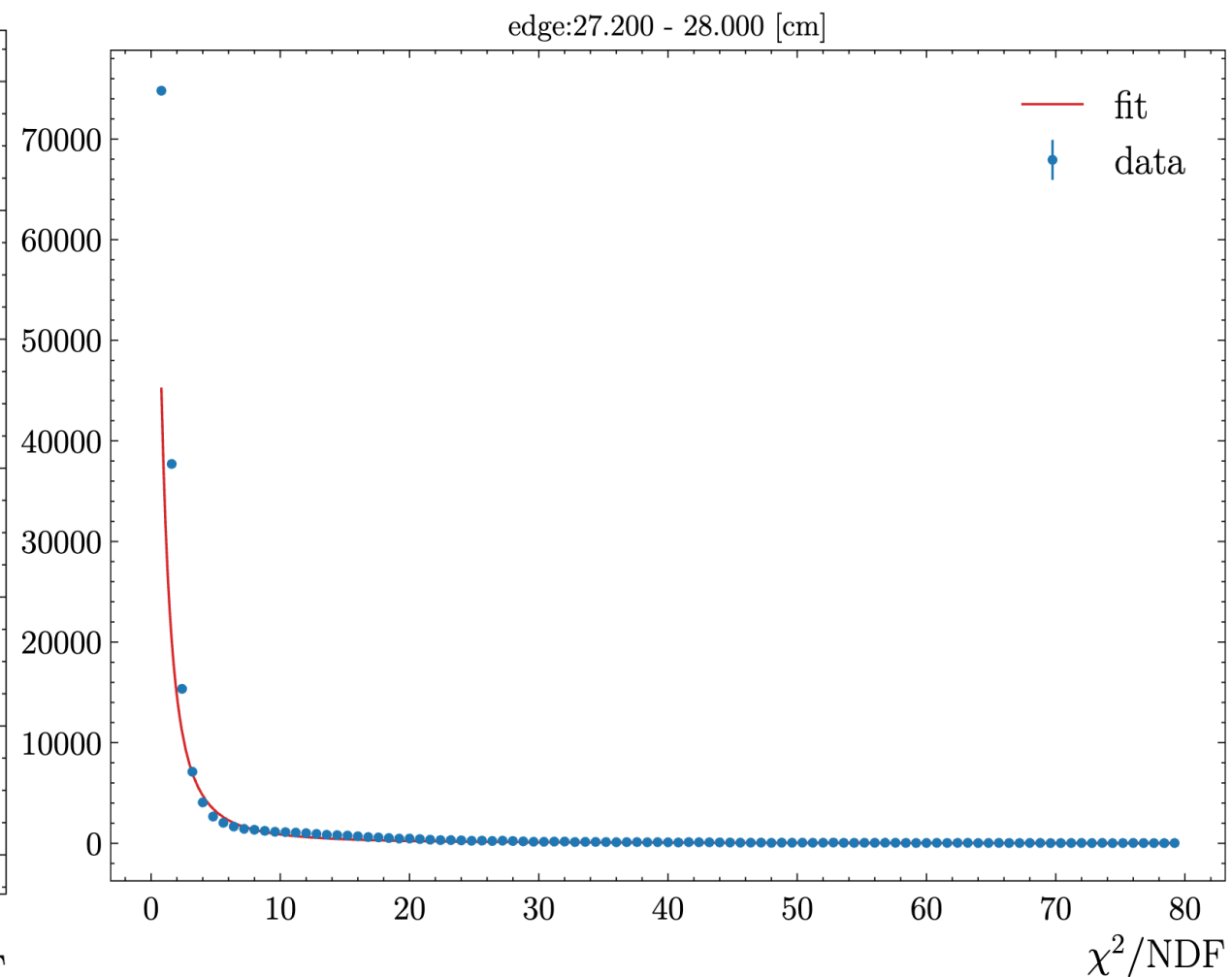
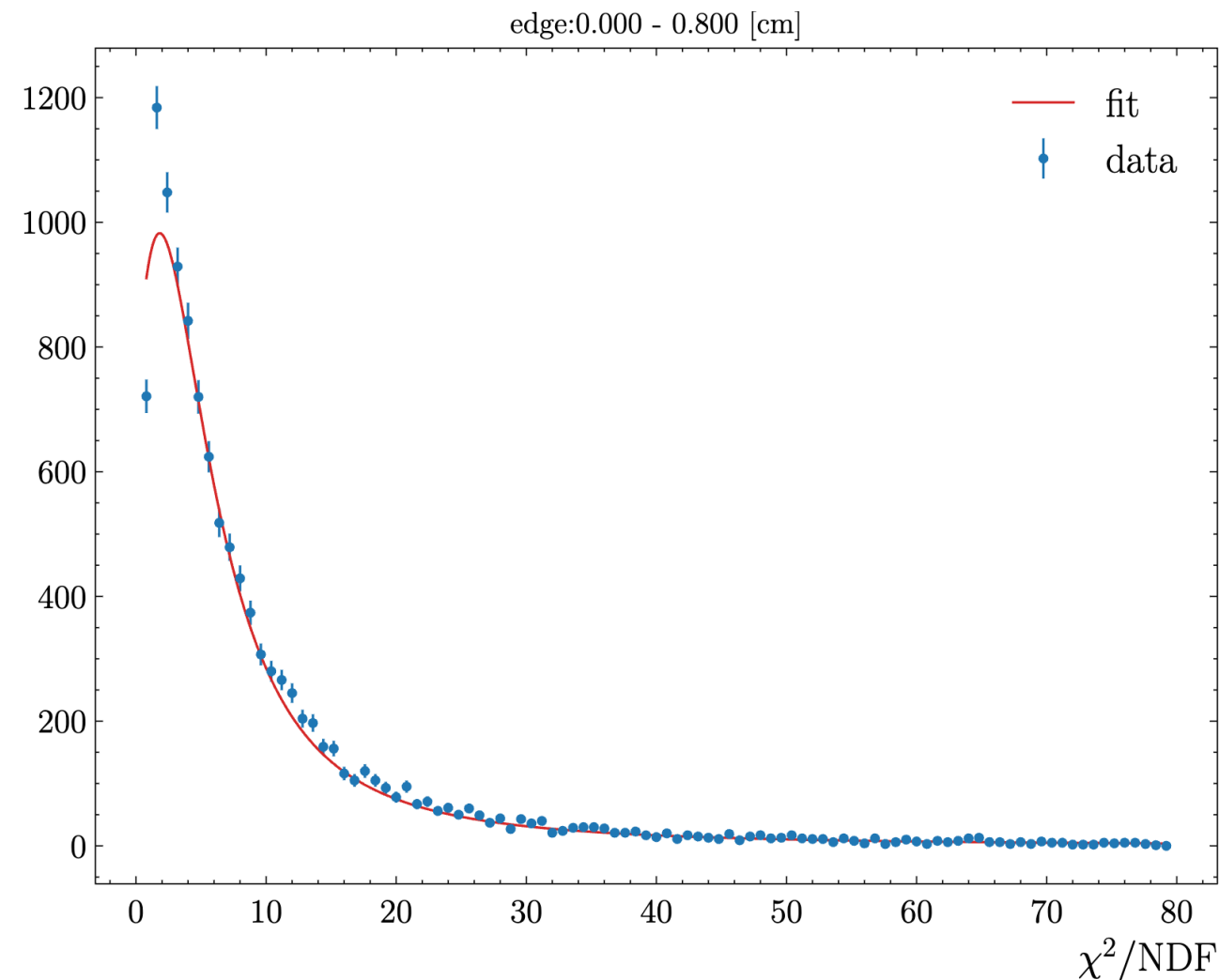
The Goal is to reproduce this plot, but sector independent:



- Sadly, the procedure is not well described in the note.
- I slice the 2D distribution along the X axis.

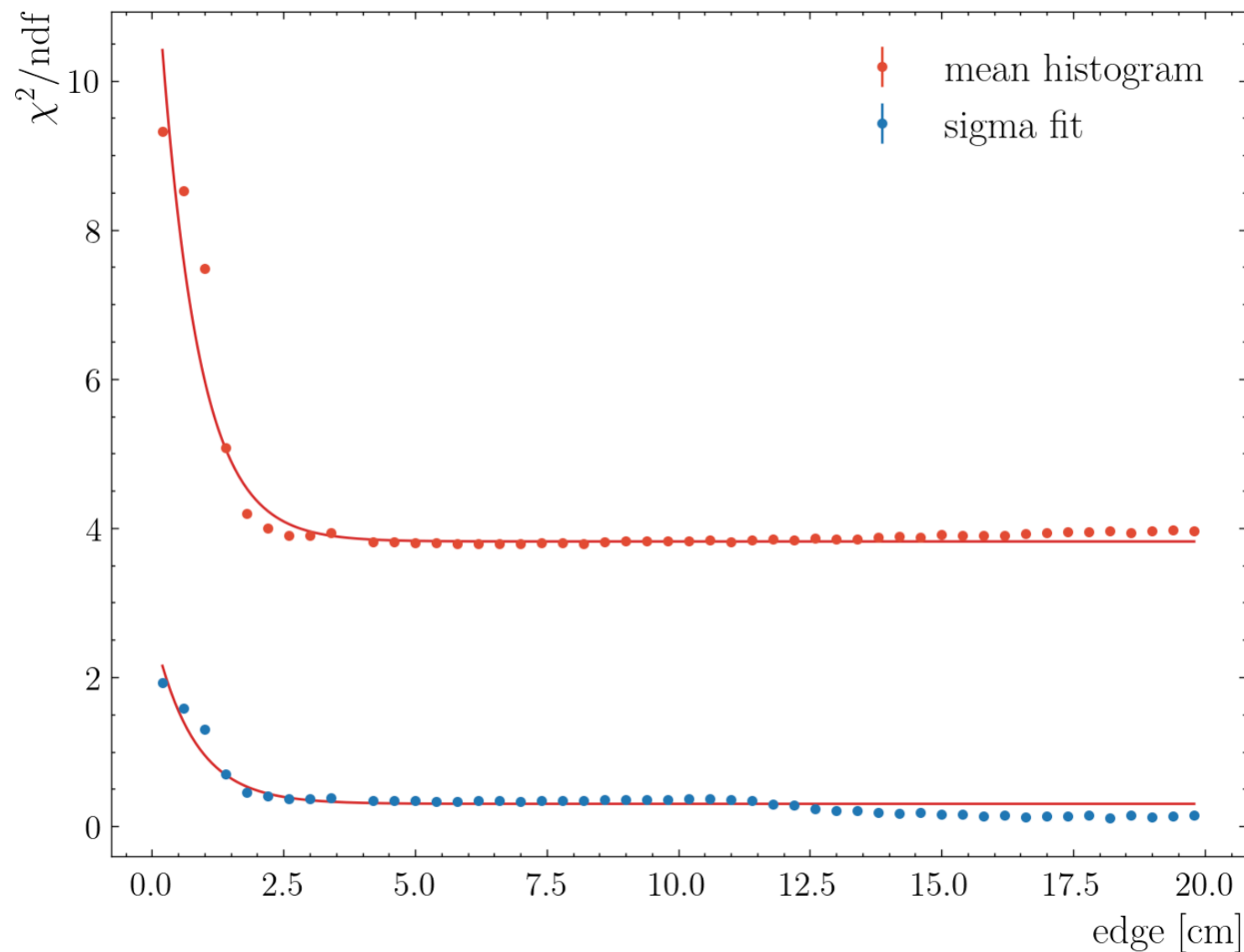


Fit with a Landau function



- Extract the mean of the slices
- Extract the sigma of the Landau
- Fit with an exponential function

$$f = A \exp(-Bx) + C$$



Task	expected	done	comment
AI network + validation	04/21	04/20	Loose some time in the cooking
Redo the background merging studies	04/25		
Improve pid, fiducials cuts	04/21		Work on the DC fiducials, need to produce a PDF with all the plots and details
Produce physics plots for pass1 review	04/30		
Finish the event mixing method	05/04		
Extract the multiplicity ratio, pT	05/15		
improve code that generated all track candidates (ALERT)	05/15		
Test the AI on real data (ALERT)	04/21		Simulation is still broken